

INFO8006 Introduction to Artificial Intelligence

Exercise session 4: Probabilistic reasoning

RECAP

Bayesian networks

With a partially observable environment, an agent reasons about the state s from its percepts e with the joint distribution $P(s, e) = P(s)P(e | s)$. When s and e are assignments of many random variables $X_1, \dots, X_n$, a *Bayesian network* represents this distribution compactly. It is a directed acyclic graph with one node per variable, each node X_i annotated with a conditional distribution $\mathbf{P}(X_i | \text{parents}(X_i))$, given as a *conditional probability table* (CPT) for discrete variables. The topology asserts the *local Markov property*, $X_i \perp \text{non-descendants}(X_i) | \text{parents}(X_i)$, hence the factorization

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i)).$$

Naive Bayes is the network in which a cause C is the only parent of each effect E_i .

d-separation

$X_i \perp X_j | Z_1, \dots, Z_m$ is guaranteed if every undirected path between X_i and X_j is *inactive*; if one path is *active*, independence is not guaranteed. A path is active iff each of its triples is active:

Triple		Active iff
Cascade	$A \rightarrow B \rightarrow C$ or $A \leftarrow B \leftarrow C$	B unobserved
Common parent	$A \leftarrow B \rightarrow C$	B unobserved
v-structure	$A \rightarrow B \leftarrow C$	B or a descendant of B observed

Given its *Markov blanket*, *i.e.* its parents, its children and the other parents of its children, a variable is independent of all the others.

Inference

Given *evidence* variables $\mathbf{E}$ observed as $\mathbf{e}$, a *query* variable Q and *hidden* variables $\mathbf{H}$, inference computes

$$\mathbf{P}(Q | \mathbf{e}) = \frac{1}{Z} \sum_{\mathbf{h}} \mathbf{P}(Q, \mathbf{h}, \mathbf{e}), \quad Z = \sum_q \sum_{\mathbf{h}} P(q, \mathbf{h}, \mathbf{e}),$$

with the joint entries written as products of CPT entries. *Inference by enumeration* evaluates the sums depth-first, in $O(n)$ space and $O(d^n)$ time for n variables with d values. *Variable elimination* evaluates them right to left and stores the intermediate *factors*, multi-dimensional arrays initialized with the CPTs given the evidence. It alternates the *join* of factors, their pointwise product over the union of their variables, and the *elimination* of a hidden variable,

summed out of the join of the factors that mention it. Its cost is that of the largest factor, which depends on the elimination ordering; finding the best ordering is NP-hard. A hidden variable H is *irrelevant*, and can be removed, unless $H \in \text{ancestors}(\{Q\} \cup \mathbf{E})$.

Parameter learning

The CPTs can be chosen in a *parameterized* family $\mathbf{P}(X \mid \theta)$, whose parameters are estimated from N i.i.d. observations $\mathbf{d} = \{x_1, \dots, x_N\}$. The *maximum likelihood estimate* (MLE) maximizes the *likelihood* $P(\mathbf{d} \mid \theta) = \prod_{j=1}^N P(x_j \mid \theta)$, or the *log-likelihood* $L(\theta) = \log P(\mathbf{d} \mid \theta)$, often by setting $\frac{\partial L}{\partial \theta} = 0$:

$$\theta_{\text{MLE}}^* = \arg \max_{\theta} P(\mathbf{d} \mid \theta).$$

Bayesian parameter learning treats θ as a random variable with a *prior* $\mathbf{P}(\theta)$, updated into the *posterior* $\mathbf{P}(\theta \mid \mathbf{d}) \propto \mathbf{P}(\mathbf{d} \mid \theta)\mathbf{P}(\theta)$. A new observation x is then predicted by the *posterior predictive* $p(x \mid \mathbf{d}) = \int p(x \mid \theta)p(\theta \mid \mathbf{d}) d\theta$, and by the *prior predictive* $p(x) = \int p(x \mid \theta)p(\theta) d\theta$ before any data. When the posterior is intractable, the *maximum a posteriori* (MAP) estimate replaces it by its mode,

$$\theta_{\text{MAP}}^* = \arg \max_{\theta} P(\theta \mid \mathbf{d}) = \arg \max_{\theta} P(\mathbf{d} \mid \theta)P(\theta).$$

Cheat sheet for Gaussian distributions

The Gaussian density of mean μ and variance σ^2 is

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^T & \mathbf{B} \end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a}, \mathbf{A}) \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b}, \mathbf{B}) \\ p(\mathbf{x} \mid \mathbf{y}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a} + \mathbf{CB}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{CB}^{-1}\mathbf{C}^T) \\ p(\mathbf{y} \mid \mathbf{x}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}). \end{aligned}$$

Conversely, if $\mathbf{x}$ and $\mathbf{y}$ have the Gaussian distributions

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{m}, \mathbf{P}) \\ p(\mathbf{y} \mid \mathbf{x}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R}), \end{aligned}$$

then their joint distribution is Gaussian with

$$p\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{m} \\ \mathbf{H}\mathbf{m} + \mathbf{u} \end{pmatrix}, \begin{pmatrix} \mathbf{P} & \mathbf{P}\mathbf{H}^T \\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R} \end{pmatrix}\right).$$

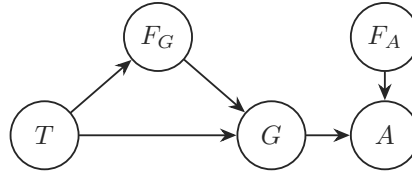
In session: exercises 1, 2.

Exercise 1 Nuclear power plant (AIMA ch. 13)

A monitoring program supervises the core of a nuclear power station. An alarm sounds when a temperature gauge, which measures the temperature of the core, exceeds a given threshold, and the program reasons about the actual temperature from the gauge and the alarm, either of which may be faulty. Consider the Boolean variables A (alarm sounds), F_A (alarm is faulty), and F_G (gauge is faulty) and the multi-valued nodes G (gauge reading) and T (actual core temperature).

- (a) Draw a Bayesian network for this domain, given that the gauge is more likely to fail when the core temperature gets too high.

Solution.



- (b) From your network topology, can you conclude $T \perp F_A \mid F_G$ and $T \perp F_A \mid F_G, A$?

Solution. The paths from T to F_A are (T, F_G, G, A, F_A) and (T, G, A, F_A) .

Knowing F_G , the triple (G, A, F_A) is inactive (v-structure, A unobserved). Since it is present in both paths, there is no active path, so $T \perp F_A \mid F_G$ is true.

For the second relation, the triple (G, A, F_A) becomes active as we observe A . The only way we can ensure conditional independence is by ensuring that at least one remaining triple is inactive in each path. For the first path, we have (T, F_G, G) which is inactive (cascade) which makes the path inactive. However, for the second path, (T, G, A) is active (cascade) which makes the path active. We have at least one active path from T to F_A which prevents us from concluding the conditional independence in this case.

- (c) Suppose there are just two possible actual and measured temperatures: low (l) and high (h). The probability that the gauge gives the correct temperature is x when it is working, but y when it is faulty. Give the conditional probability table associated with G .

Solution. The CPT for G is shown below. Note the semantics of F_G , which is true when the gauge is faulty, *i.e.* not working properly.

T	F_G	$P(G = l \mid T, F_G)$
l	0	x
l	1	y
h	0	$1 - x$
h	1	$1 - y$

- (d) Suppose the alarm is always triggered by high measured temperatures, unless it is faulty, in which case it never sounds. Give the conditional probability table associated with A .

Solution.

G	F_A	$P(A = 1 \mid G, F_A)$
l	0	0
l	1	0
h	0	1
h	1	0

- (e) Suppose the gauge is not faulty and the alarm is triggered. Calculate an expression for the probability that the temperature of the core is too high, in terms of the various conditional probabilities in the network.

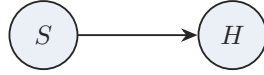
Solution. The probability of interest here is $P(T = h \mid A = 1, F_G = 0)$. Because the alarm's behavior is deterministic, we can deduce that G indicates high (h). Hence, $P(T = h \mid A = 1, F_G = 0) = P(T = h \mid A = 1, G = h, F_G = 0)$. We also see that $A \perp T \mid G$. Therefore, we only need to calculate

$$\begin{aligned} P(T = h \mid G = h, F_G = 0) &= \frac{P(T = h, G = h, F_G = 0)}{P(G = h, F_G = 0)} \\ &= \frac{P(G = h \mid T = h, F_G = 0)P(F_G = 0 \mid T = h)P(T = h)}{\sum_t P(G = h \mid t, F_G = 0)P(F_G = 0 \mid t)P(t)}, \end{aligned}$$

which we cannot develop more as we don't know $P(T)$ and $P(F_G \mid T)$.

Exercise 2 Energy of an inference server

A scheduling program manages a data center in which a GPU server answers the requests sent to a language model. To plan the power supply, the scheduler models the energy the server consumes as a function of its load, with the following network



where S is a random variable denoting the number of requests the server receives in an hour, and H a random variable denoting the energy, in kilowatt-hours, that it consumes during that hour. The scheduler chooses the following relations for each variable

$$P(S = s) = \text{Poisson}(S = s; \lambda) = \frac{\lambda^s e^{-\lambda}}{s!} \quad (1)$$

$$H = \omega S + H_0 + \mathcal{N}(0, \sigma^2) \quad (2)$$

where H_0 is the energy the server consumes when idle, which is known, and ω is the energy consumed per request.

- (a) Identify the parameters of the Bayesian network and give the joint log-likelihood of one pair (s_i, h_i) given the model.

Solution. Given the topology of the network, we have to identify $P(S = s)$ and $P(H = h \mid S = s)$. The former is given by (1) whereas the latter can be deduced from (2). We have

$$P(S = s) = \text{Poisson}(S = s; \lambda)$$

and

$$p(h \mid s) = \mathcal{N}(h \mid \omega s + H_0, \sigma^2).$$

The parameters of those distributions are $\theta = [\lambda, \omega, \sigma]$.

The likelihood of a pair (s_i, h_i) can be derived from the network. We have

$$\begin{aligned} P((s_i, h_i) \mid \theta) &= P(s_i)P(h_i \mid s_i) \\ &= \frac{\lambda^{s_i} e^{-\lambda}}{s_i!} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(h_i - \omega s_i - H_0)^2}{2\sigma^2}} \end{aligned}$$

from which we derive the log-likelihood

$$L((s_i, h_i); \theta) = s_i \log \lambda - \lambda - \log s_i! - \log \sqrt{2\pi}\sigma - \frac{(h_i - \omega s_i - H_0)^2}{2\sigma^2}. \quad (3)$$

- (b) The scheduler collects a set $\mathbf{d} = \{(s_i, h_i)\}_{i=1}^N$ of independent observations. Derive the maximum likelihood estimate of the parameters.

Solution. We first have to define the joint likelihood of those observations. Since they are assumed i.i.d., we have

$$P(\mathbf{d} | \theta) = \prod_{i=1}^N P((s_i, h_i) | \theta)$$

Hence, deriving the log-likelihood

$$\begin{aligned} L(\mathbf{d}; \theta) &= \sum_i L((s_i, h_i); \theta) \\ &= -N(\lambda + \log \sqrt{2\pi}\sigma) + \sum_i s_i \log \lambda - \log s_i! - \frac{(h_i - \omega s_i - H_0)^2}{2\sigma^2} \end{aligned}$$

The maximum likelihood parameters can be found by cancelling the derivative of the log-likelihood w.r.t. each parameter.

For λ we have

$$\begin{aligned} \frac{\partial L}{\partial \lambda} &= -N + \sum_i \frac{s_i}{\lambda} \\ &= 0 \end{aligned}$$

which gives $\lambda_{\text{MLE}} = \frac{1}{N} \sum_i s_i$.

For σ we have

$$\begin{aligned} \frac{\partial L}{\partial \sigma} &= -\frac{1}{\sigma} \left(N - \sum_i \frac{(h_i - \omega s_i - H_0)^2}{\sigma^2} \right) \\ &\propto N - \frac{1}{\sigma^2} \sum_i (h_i - \omega s_i - H_0)^2 \\ &= 0 \end{aligned}$$

which gives $\sigma^2_{\text{MLE}} = \frac{1}{N} \sum_i (h_i - \omega s_i - H_0)^2$.

For ω we have

$$\begin{aligned} \frac{\partial L}{\partial \omega} &= \sum_i \frac{s_i (h_i - \omega s_i - H_0)}{\sigma^2} \\ &\propto \left(\sum_i s_i (h_i - H_0) \right) - \omega \sum_i s_i^2 \\ &= 0 \end{aligned}$$

which gives $\omega_{\text{MLE}} = \frac{\sum_i s_i (h_i - H_0)}{\sum_i s_i^2}$.

- (c) The scheduler has just started to collect observations, and maximum likelihood can overfit when they are few. It therefore treats the parameter λ , about which it has no strong knowledge, as uncertain. Using a prior $\lambda \sim \text{Gamma}(\alpha, \beta)$, what would be the posterior distribution $p(\lambda \mid \mathbf{s})$, with $\mathbf{s} = \{s_i\}_{i=1}^N$?

Hint: $\text{Gamma}(x \mid \alpha, \beta) = \frac{1}{Z} x^{\alpha-1} e^{-\beta x}$, with Z a normalizing constant.

Solution. Using Bayes' rule

$$\begin{aligned} p(\lambda \mid \mathbf{s}) &\propto P(\mathbf{s} \mid \lambda) p(\lambda) \\ &\propto \lambda^{\alpha-1} e^{-\beta \lambda} \prod_i \frac{\lambda^{s_i} e^{-\lambda}}{s_i!} \\ &= \lambda^{\alpha-1} e^{-\beta \lambda} \left(\frac{\lambda^{\sum_i s_i} e^{-N\lambda}}{\prod_i s_i!} \right) \\ &\propto \lambda^{(\alpha + \sum_i s_i - 1)} e^{-(\beta + N)\lambda} \end{aligned}$$

which corresponds to $\text{Gamma}(\alpha + \sum_i s_i, \beta + N)$, the posterior density. Note that $\propto$ stands for the normalizing constant of the posterior density. Everything that does not depend explicitly on the parameter λ can be absorbed in the constant if it multiplies the whole.

- (d) Knowing that the mean, the variance and the mode of $\text{Gamma}(\alpha, \beta)$ are respectively $\frac{\alpha}{\beta}$, $\frac{\alpha}{\beta^2}$ and $\frac{\alpha-1}{\beta}$, interpret how they evolve w.r.t. N between the prior and the posterior.

Solution. We have

$$\begin{aligned} \mathbb{E}(\lambda) : \frac{\alpha}{\beta} &\Rightarrow \frac{\alpha + \sum_i s_i}{\beta + N}, \\ \mathbb{V}(\lambda) : \frac{\alpha}{\beta^2} &\Rightarrow \frac{\alpha + \sum_i s_i}{(\beta + N)^2} \end{aligned}$$

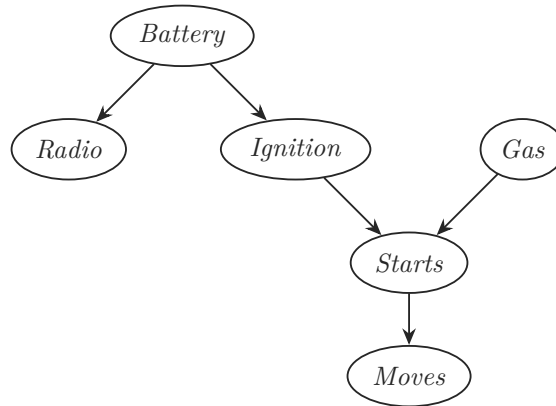
and

$$\arg \max_{\lambda} p(\lambda \mid \mathbf{s}) : \frac{\alpha-1}{\beta} \Rightarrow \frac{\alpha + \sum_i s_i - 1}{\beta + N}.$$

When $N = 0$, the prior and the posterior are similar, which is expected. When $N \rightarrow \infty$, knowing that $s \in \mathbb{N}$ by definition (a number of requests), we can assume that, for reasonable values of α, β , we have $\sum_i s_i \gg \alpha$ and $N \gg \beta$. This implies that the mean and the mode converge to the same value corresponding to λ_{MLE} . The variance decreases towards 0, which suggests that λ_{MLE} is the best and unique estimate of λ for an infinite amount of observations. This illustrates the trade-off between likelihood and prior in a Bayesian parameter learning setting. As more data are observed, the uncertainty about a parameter decreases smoothly from the prior to the likelihood.

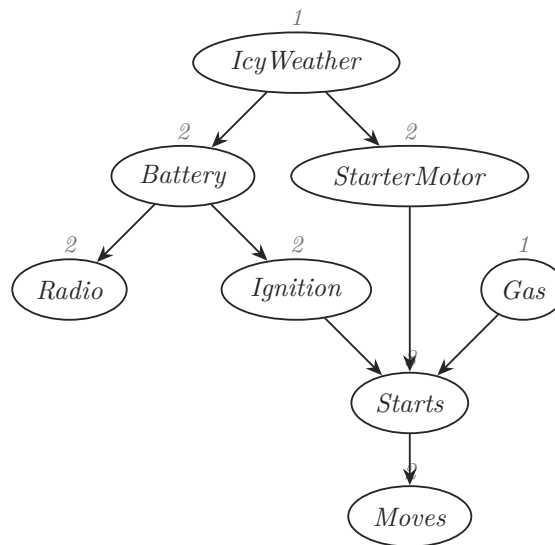
Exercise 3 Car diagnosis (AIMA ch. 13)

Consider the following Bayesian network, which describes some features of a car's electrical system and engine. Each variable is Boolean, and the true value indicates that the corresponding aspect of the vehicle is in working order.



- (a) Extend the network with the Boolean variables IcyWeather and StarterMotor.

Solution. IcyWeather is not caused by any of the car-related variables, so needs no parents. It directly affects the battery and the starter motor. StarterMotor is an additional precondition for Starts.



- (b) According to your knowledge of cars, give reasonable conditional probability tables for all the nodes.

Solution. Reasonable probabilities may vary a lot depending on the kind of car and perhaps the personal experience of the assessor. The following values indicate the general order of magnitude and relative values that would be reasonable:

- A reasonable prior for IcyWeather might be 0.05 (depending on location and season).
- $P(\text{Battery} \mid \text{IcyWeather}) = 0.95, P(\text{Battery} \mid \neg \text{IcyWeather}) = 0.997$.
- $P(\text{StarterMotor} \mid \text{IcyWeather}) = 0.98, P(\text{StarterMotor} \mid \neg \text{IcyWeather}) = 0.999$.
- $P(\text{Radio} \mid \text{Battery}) = 0.9999, P(\text{Radio} \mid \neg \text{Battery}) = 0.05$.
- $P(\text{Ignition} \mid \text{Battery}) = 0.998, P(\text{Ignition} \mid \neg \text{Battery}) = 0.01$.
- $P(\text{Gas}) = 0.995$.
- $P(\text{Starts} \mid \text{Ignition}, \text{StarterMotor}, \text{Gas}) = 0.9999$, other entries 0.0.
- $P(\text{Moves} \mid \text{Starts}) = 0.998, P(\text{Moves} \mid \neg \text{Starts}) = 0$.

- (c) How many independent values are contained in the joint probability distribution for eight Boolean nodes, assuming that no conditional independence relations are known to hold among them?

Solution. With 8 Boolean variables, the joint has $2^8 - 1 = 255$ independent entries.

- (d) How many independent probability values do your network tables contain?

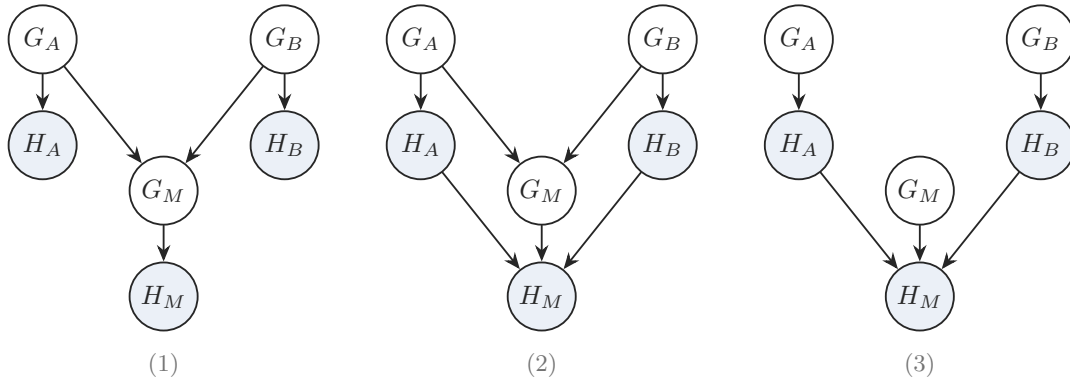
Solution. Given the topology with IcyWeather and StarterMotor, the total number of independent CPT entries is $1 + 2 + 2 + 2 + 2 + 1 + 8 + 2 = 20$.

Exercise 4 Evolutionary model merging (AIMA ch. 13)

Evolutionary model merging builds a new language model M from two parent models A and B : each component of M is copied from one of the parents. Consider one component, and let G_x be the language it was mostly trained on in model x , with possible values e (English) or f (French). A common *hypothesis* is that the default language of a model follows from this component by a simple mechanism: the language H_x in which model x answers by default is the same as G_x with some probability s .

Furthermore, M copies the component from either parent with equal probability, and a random perturbation flips it with a small nonzero probability m .

- (a) Which of the following networks claim that $\mathbf{P}(G_M \mid G_A, G_B) = \mathbf{P}(G_M)$? Shaded variables are observed.



Solution. The third one, as the node G_M does not have any parents.

- (b) Which of the networks make independence claims that are consistent with the hypothesis about the default language?

Solution. The first and second ones. The second one is also valid as it makes *fewer* assumptions about the dependencies between the variables, *i.e.* it is more general than the first one.

- (c) Write down the CPT for the G_M node in the first network, in terms of s and m .

Solution.

g_A	g_B	$P(G_M = f \mid g_A, g_B)$
f	f	$1 - m$
f	e	0.5
e	f	0.5
e	e	m

The CPT does not depend on s , which only relates G_M and H_M .

- (d) Suppose that $P(G_A = f) = P(G_B = f) = q$. In the first network, derive an expression for $P(G_M = f)$ in terms of s , m and q .

Solution.

$$\begin{aligned}
 P(G_M = f) &= \sum_{g_A} \sum_{g_B} P(G_M = f, g_A, g_B) \\
 &= \sum_{g_A} \sum_{g_B} P(G_M = f \mid g_A, g_B) P(g_A) P(g_B) \\
 &= (1 - m)q^2 + 0.5(1 - q)q + 0.5q(1 - q) + m(1 - q)^2 \\
 &= q + m - 2qm
 \end{aligned}$$

Again, s does not appear.

- (e) Over many generations of merges, we expect the distribution of the component to be the same in the parents and in the merged models. Use this to calculate the value of q . Only about 10 % of the merged models answer in French by default: explain why the hypothesis described at the beginning of this question must be wrong.

Solution. At equilibrium, $P(G_M = f) = P(G_A = f) = q$, so $q = q + m - 2qm$, *i.e.* $m(1 - 2q) = 0$. Since $m > 0$, $q = 0.5$. Then $P(H_M = f) = sq + (1 - s)(1 - q) = 0.5$ whatever s : half of the merged models should answer in French, against about 10 % observed, which invalidates the hypothesis.

Exercise 5 Gaussian parameter learning with known σ^2

You have in your possession a high-tech laser that emits light of intensity μ_l with great precision. You want to use it to test a new light-intensity measurement device. You assume that your measurements y_i follow a Gaussian distribution $\mathcal{N}(\mu_y, \sigma^2)$ with $\mu_y \sim \mathcal{N}(\mu_l, \sigma_l^2)$ corresponding to the model of your laser given in the manual. You assume that σ^2 is fixed and that you can determine it.

You have access to a calibration set $\mathcal{Y}_c$ of C observations, and you want to predict a new measurement y of your device. You want to study how much $\mathcal{Y}_c$ reduces your uncertainty about this measurement.

- (a) Compute the prior and posterior predictive distributions $P(y)$ and $P(y \mid \mathcal{Y}_c)$ if you have access to one observation, *i.e.* $\mathcal{Y}_c = \{y_o\}$.

Solution. The following exercise uses the Gaussian model identities (see the cheat sheet of the recap).

We first identify the components we have:

- $p(y \mid \mu_y) = \mathcal{N}(y \mid \mu_y, \sigma^2)$
- $p(\mu_y) = \mathcal{N}(\mu_y \mid \mu_l, \sigma_l^2)$

Prior predictive

$$P(y) = \int P(y \mid \mu_y) P(\mu_y) d\mu_y.$$

From Gaussian identities, you can identify different parameters as follows

$$\begin{aligned}
 \mathbf{x} &= \mu_y \\
 \mathbf{m} &= \mu_l \\
 \mathbf{P} &= \sigma_l^2 \\
 \mathbf{y} &= y \\
 \mathbf{H} &= 1 \\
 \mathbf{u} &= 0 \\
 \mathbf{R} &= \sigma^2
 \end{aligned}$$

which corresponds to the joint

$$P(\mu_y, y) = \mathcal{N} \left(\begin{pmatrix} \mu_l \\ \mu_l \end{pmatrix}, \begin{pmatrix} \sigma_l^2 & \sigma_l^2 \\ \sigma_l^2 & \sigma_l^2 + \sigma^2 \end{pmatrix} \right)$$

from which we extract $p(y) = \mathcal{N}(y \mid \mu_l, \sigma^2 + \sigma_l^2)$.

Posterior predictive

$$P(y \mid \mathcal{Y}_c) = \int P(y \mid \mu_y) P(\mu_y \mid \mathcal{Y}_c) d\mu_y.$$

We first have to identify the posterior $P(\mu_y \mid \mathcal{Y}_c)$. Once again, from Gaussian identities, knowing $P(\mathcal{Y}_c \mid \mu_y)$ and $P(\mu_y)$ you can identify different parameters as follows

$$\begin{aligned} \mathbf{x} &= \mu_y \\ \mathbf{m} &= \mu_l \\ \mathbf{P} &= \sigma_l^2 \\ \mathbf{y} &= y_o \\ \mathbf{H} &= 1 \\ \mathbf{u} &= 0 \\ \mathbf{R} &= \sigma^2 \end{aligned}$$

which corresponds to the joint

$$P(\mu_y, y_o) = \mathcal{N} \left(\begin{pmatrix} \mu_l \\ \mu_l \end{pmatrix}, \begin{pmatrix} \sigma_l^2 & \sigma_l^2 \\ \sigma_l^2 & \sigma_l^2 + \sigma^2 \end{pmatrix} \right)$$

from which we extract

$$\begin{aligned} p(\mu_y \mid y_o) &= \mathcal{N}(\mu_y \mid \mu_{\text{post}}, \sigma_{\text{post}}^2) \\ &= \mathcal{N} \left(\mu_y \mid \mu_l + \frac{\sigma_l^2}{\sigma^2 + \sigma_l^2} (y_o - \mu_l), \sigma_l^2 - \frac{\sigma_l^4}{\sigma^2 + \sigma_l^2} \right). \end{aligned}$$

After simplification, we identify the posterior parameters as

$$\mu_{\text{post}} = \frac{\sigma^2}{\sigma^2 + \sigma_l^2} \mu_l + \frac{\sigma_l^2}{\sigma^2 + \sigma_l^2} y_o$$

and

$$\sigma_{\text{post}}^2 = \frac{\sigma^2}{\sigma_l^2 + \sigma^2} \sigma_l^2.$$

Following the same development as for the prior predictive, we have

$$p(y \mid \mathcal{Y}_c) = \mathcal{N}(y \mid \mu_{\text{post}}, \sigma^2 + \sigma_{\text{post}}^2).$$

- (b) Looking at your predictive uncertainty, what is the benefit of incorporating $\mathcal{Y}_c$?

Solution. As a reminder, we found

$$p(y) = \mathcal{N}(y \mid \mu_l, \sigma^2 + \sigma_l^2).$$

$$p(y \mid \mathcal{Y}_c) = \mathcal{N}(y \mid \mu_{\text{post}}, \sigma^2 + \sigma_{\text{post}}^2).$$

Comparing the variances, since $\sigma^2/(\sigma^2 + \sigma_l^2) \leq 1$,

$$\sigma_{\text{post}}^2 = \frac{\sigma^2}{\sigma^2 + \sigma_l^2} \sigma_l^2 \leq \sigma_l^2, \quad \text{hence} \quad \sigma^2 + \sigma_{\text{post}}^2 \leq \sigma^2 + \sigma_l^2 :$$

the posterior predictive has a smaller or equal variance than the prior predictive. Incorporating the calibration set reduces the uncertainty about the measurement.

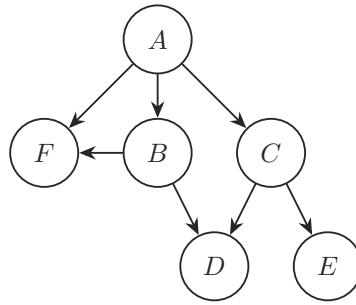
If we computed the variance for a calibration set of C observations, we would have found a posterior variance

$$\sigma_{\text{post}}^2 = \frac{\sigma^2}{C\sigma_l^2 + \sigma^2} \sigma_l^2.$$

Since σ^2 is fixed, including more observations reduces the posterior predictive variance further. In the limit case $C \rightarrow \infty$, the posterior predictive variance converges to σ^2 which corresponds to the uncertainty of the instrument (you can't reach a lower uncertainty, it is called the aleatoric uncertainty).

Exercise 6 Independence and variable elimination

Considering the Bayesian network below, which of the following statements are enforced by the network structure?



Solution. We apply the d-separation algorithm. To show that two variables X and Y could be dependent, it is sufficient to find a single *active* (undirected) path from X to Y . A path is active if all of its consecutive triples are active.

- (a) $\mathbf{P}(B, C) = \mathbf{P}(B)\mathbf{P}(C)$

Solution. This is true iff (if and only if) $B \perp C$ (B is independent from C). The three paths that link B and C are (B, A, C) , (B, D, C) and (B, F, A, C) . (B, A, C) is active because A is unobserved (common parent). We cannot guarantee that $B \perp C$.

- (b) $\mathbf{P}(B, C | A) = \mathbf{P}(B | A)\mathbf{P}(C | A)$

Solution. This is true iff $B \perp C | A$. This time, (B, A, C) is not active because A is known. (B, D, C) is not active either because D is unobserved (v-structure). In (B, F, A, C) , the triple (B, F, A) is not active (v-structure). (F, A, C) is not active either. We can guarantee that $B \perp C | A$.

- (c) $\mathbf{P}(F, E) = \mathbf{P}(F)\mathbf{P}(E)$

Solution. This is true iff $F \perp E$. The four paths that link F and E are (F, A, C, E) , (F, A, B, D, C, E) , (F, B, A, C, E) and (F, B, D, C, E) . In (F, A, C, E) , the triples (F, A, C) (common parent) and (A, C, E) (cascade) are both active. We cannot guarantee that $F \perp E$.

(d) $\mathbf{P}(F \mid A, D, E) = \mathbf{P}(F \mid A, D)$

Solution. This is true iff $F \perp E \mid A, D$. This time, the triples (F, A, C) , (F, A, B) and (B, A, C) are inactive (common parent, A observed). In (F, B, D, C, E) , (F, B, D) is active. Indeed, observing an extremity of a triple does not change whether it is active: only the center variable matters, and, for a v-structure, its descendants. Then we have (B, D, C) which is active (v-structure, D observed) and (D, C, E) also. Hence, (F, B, D, C, E) is active and we cannot guarantee that $F \perp E \mid A, D$.

(e) $\mathbf{P}(B, E) = \sum_{a,c,d,f} P(a)\mathbf{P}(B \mid a)P(c \mid a)\mathbf{P}(d \mid B, c)\mathbf{P}(E \mid c)\mathbf{P}(f \mid a, B)$

Solution. We notice that

$$\mathbf{P}(A, B, C, D, E, F) = \mathbf{P}(A)\mathbf{P}(B \mid A)\mathbf{P}(C \mid A)\mathbf{P}(D \mid B, C)\mathbf{P}(E \mid C)\mathbf{P}(F \mid A, B)$$

corresponds exactly to

$$\mathbf{P}(X_1, \dots, X_n) = \prod_{i=1}^n \mathbf{P}(X_i \mid \text{parents}(X_i)),$$

for the considered network, which we know is guaranteed. We also know that

$$\mathbf{P}(B, E) = \sum_{a,c,d,f} \mathbf{P}(a, B, c, d, E, f)$$

Hence, the initial statement is always true.

For the same network, use inference by variable elimination to compute $\mathbf{P}(E \mid A = 1, B = 1)$.

Solution. The hidden variables are C, D and F . By the relevance theorem, D and F , which are not ancestors of $\{E, A, B\}$, could be removed from the network; we keep them to see why. We have

$$\begin{aligned} \mathbf{P}(E \mid A = 1, B = 1) &= \frac{1}{Z} \sum_{c,d,f} \mathbf{P}(A = 1, B = 1, c, d, E, f) \\ &= \frac{1}{Z} \sum_{c,d,f} P(A = 1)P(B = 1 \mid A = 1)P(c \mid A = 1) \\ &\quad P(d \mid B = 1, c)\mathbf{P}(E \mid c)P(f \mid A = 1, B = 1) \\ &= \frac{1}{Z} P(A = 1)P(B = 1 \mid A = 1) \sum_f P(f \mid A = 1, B = 1) \\ &\quad \sum_c P(c \mid A = 1)\mathbf{P}(E \mid c) \sum_d P(d \mid B = 1, c). \end{aligned}$$

We define the initial factors as

$$\begin{aligned} f_1 &= P(A = 1) \\ f_2 &= P(B = 1 \mid A = 1) \\ f_3(C) &= P(C \mid A = 1) \\ f_4(E, C) &= P(E \mid C) \\ f_5(C, D) &= P(D \mid B = 1, C) \\ f_6(F) &= P(F \mid A = 1, B = 1) \end{aligned}$$

and the composite factors as

$$\begin{aligned}
f_7 &= \sum_f f_6(f) = 1 \\
f_8(C) &= \sum_d f_5(C, d) = 1 \\
f_9(C, E) &= f_3(C) \times f_4(E, C) \times f_8(C) \\
f_{10}(E) &= \sum_c f_9(c, E) \\
f_{11}(E) &= f_1 \times f_2 \times f_7 \times f_{10}(E).
\end{aligned}$$

Finally,

$$\mathbf{P}(E \mid A = 1, B = 1) = \frac{1}{Z} f_{11}(E) = \frac{f_{11}(E)}{\sum_e f_{11}(e)}.$$

Eliminating F and D gives factors equal to 1, as the relevance theorem predicts, and the constant $f_1 f_2$ cancels in the normalization. What remains is

$$\mathbf{P}(E \mid A = 1, B = 1) = \sum_c P(c \mid A = 1) \mathbf{P}(E \mid c) = \mathbf{P}(E \mid A = 1),$$

already normalized. The evidence $B = 1$ does not matter: $E \perp B \mid A$ by d-separation, since every path from B to E reaches C through (B, A, C) , (B, D, C) or (B, F, A, C) , all inactive given A , as in (b).

Exercise 7 Recognizing Pacman (CS188, Spring 2014)

The ghosts of the maze are controlled by a program that watches the maze through cameras and must recognize Pacman in their pictures. It learns to do so from a set of pictures of Pacman, and of ghosts as negative examples.

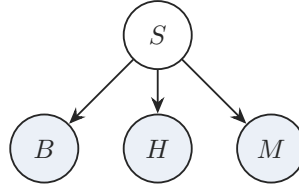


Because the pictures are black and white, and were taken from various angles, the program identifies Pacman from salient features: the presence of a bowtie B , hat H or mustache M . The following table summarizes the content of the pictures. Each feature takes realization in $\{0, 1\}$, where 0 and 1 mean the feature is respectively absent and present. The subject of the picture is described by a random variable $S \in \{0, 1\}$, where 0 is a ghost and 1 is Pacman.

B	H	M	S
0	0	0	1
1	0	0	0
1	0	1	1
1	1	0	0
0	1	0	0
1	1	1	1

- (a) Suppose the program is a naive Bayes classifier. Draw the Bayesian network that represents the dependencies between S , B , H and M for the program.

Solution.



- (b) Write the Bayesian classification rule for this problem, *i.e.* the formula that given a data point (b, h, m) returns the most likely subject. Write the formula in terms of conditional and prior probabilities. What does the formula become under the assumptions of the program?

Solution. Given (b, h, m) , the most likely subject is the one of highest posterior probability,

$$\begin{aligned} s^* &= \arg \max_s P(s \mid b, h, m) \\ &= \arg \max_s P(b, h, m \mid s)P(s). \end{aligned}$$

Under the naive Bayes assumptions of the program, B , H and M become independent conditionally to S , *i.e.* $\mathbf{P}(B, H, M \mid S) = \mathbf{P}(B \mid S)\mathbf{P}(H \mid S)\mathbf{P}(M \mid S)$. Then, the formula becomes

$$s^* = \arg \max_s P(b \mid s)P(h \mid s)P(m \mid s)P(s).$$

- (c) What are the parameters of this model? Give estimates of these parameters according to the pictures of the table.

Solution. The parameters of the model are the elements of the prior vector $\mathbf{P}(S)$ and the (conditional) probability matrices $\mathbf{P}(B \mid S)$, $\mathbf{P}(H \mid S)$ and $\mathbf{P}(M \mid S)$. Their maximum likelihood estimates are the relative frequencies of the corresponding events among the pictures.

S	$P(S)$	$P(B = 1 \mid S)$	$P(H = 1 \mid S)$	$P(M = 1 \mid S)$
0	$\frac{3}{6}$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{0}{3}$
1	$\frac{3}{6}$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{2}{3}$

The estimate $P(M = 1 \mid S = 0) = 0$ is a problem: no ghost of the six pictures has a mustache, so any picture with a mustache gets the posterior probability 0 of being a ghost, whatever its other features. With so few pictures, a prior on the parameters helps. With a Beta(2, 2) prior on each, the MAP estimates are $(k + 1)/(n + 2)$ for k occurrences among n pictures (*add-one smoothing*), *e.g.* $P(M = 1 \mid S = 0) = 1/5$.

- (d) Pacman eventually shows up in front of a camera wearing a bowtie, but no hat or mustache. Will the program recognize him?

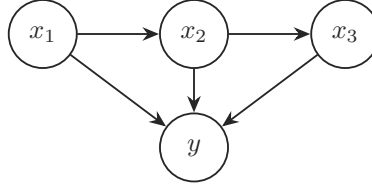
Solution. The program will recognize Pacman if $s^* = 1$ for $(b, h, m) = (1, 0, 0)$. Using the parameters estimated previously, we have

$$\begin{aligned} P(b \mid 0)P(h \mid 0)P(m \mid 0)P(0) &= \frac{2}{3} \times \left(1 - \frac{2}{3}\right) \times \left(1 - \frac{0}{3}\right) \times \frac{3}{6} \approx 0.111 \\ P(b \mid 1)P(h \mid 1)P(m \mid 1)P(1) &= \frac{2}{3} \times \left(1 - \frac{1}{3}\right) \times \left(1 - \frac{2}{3}\right) \times \frac{3}{6} \approx 0.074. \end{aligned}$$

Therefore, $s^* = 0$, meaning that the program will *not* recognize Pacman, who escapes.

Exercise 8 Predict the leaderboard score

A leaderboard ranks language models by an overall score. The Bayesian network below represents how this score is computed. In this model, x_1 , x_2 and x_3 respectively denote the scores obtained by a model on a validation set, on a public benchmark and on a held-out test set. The three evaluations share the same harness, and a model that does well on one tends to do well on the next. In particular, $x_2 \sim \mathcal{N}(a_1x_1 + \mu_2, \sigma_2^2)$ and $x_3 \sim \mathcal{N}(a_2x_2 + \mu_3, \sigma_3^2)$. Finally, $y \sim \mathcal{N}(a_3x_1 + a_4x_2 + a_5x_3 + \mu_y, \sigma_y^2)$ stands for the overall score, which is a linear combination of the three scores plus some Gaussian noise due to rounding errors. Answer the following questions about this model.



- (a) Assuming the parameters of the model are known, what is the expected value of y given x_1 and x_2 ?

Solution. Our task is to find the expectation

$$\mathbb{E}_{p(y|x_1, x_2)}[y] = \int y p(y | x_1, x_2) dy.$$

We know that

$$p(y | x_1, x_2) = \int p(y | x_1, x_2, x_3) p(x_3 | x_1, x_2) dx_3,$$

where $p(y | x_1, x_2, x_3)$ is given in the statement and $p(x_3 | x_1, x_2) = p(x_3 | x_2)$, since $x_3 \perp x_1 | x_2$. Both are linear Gaussian distributions. Therefore, we have

$$p(y | x_1, x_2) = \mathcal{N}\left(y | a_3x_1 + a_4x_2 + a_5(a_2x_2 + \mu_3) + \mu_y, (a_5\sigma_3)^2 + \sigma_y^2\right)$$

and, by definition of a Gaussian distribution,

$$\mathbb{E}_{p(y|x_1, x_2)}[y] = a_3x_1 + (a_4 + a_5a_2)x_2 + a_5\mu_3 + \mu_y.$$

- (b) Suppose now that the parameters are unknown, and that the scores x_3 on the held-out test set are not observed. Given a learning set $d = \{(x_{i,1}, x_{i,2}, y_i)\}$ of N independent and identically distributed points, derive the maximum likelihood estimate of the parameters of $p(y | x_1, x_2)$.

Solution. We know that the distribution of y given x_1 and x_2 takes the form $\mathcal{N}(y | w_1x_1 + w_2x_2 + b, \sigma^2)$. Then, our task is to find the parameters $h = (w_1, w_2, b, \sigma)$ that maximize the likelihood of d , i.e. the *maximum likelihood estimation* (MLE)

$$\begin{aligned} h_{\text{MLE}} &= \arg \max_h p(d | h) \\ &= \arg \max_h \prod_i p(x_i, y_i | h) \\ &= \arg \max_h \log \prod_i p(x_i, y_i | h) \\ &= \arg \max_h \sum_i \log p(x_i, y_i | h) \\ &= \arg \max_h \sum_i \log p(y_i | h, x_i) + \log p(x_i) \end{aligned}$$

$$\begin{aligned}
&= \arg \max_h \sum_i \log p(y_i | h, x_i) \\
&= \arg \max_h \sum_i \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(w^T x_i - y_i)^2}{2\sigma^2} \right) \right] \\
&= \arg \max_h \sum_i -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(w^T x_i - y_i)^2}{2\sigma^2} \\
&= \arg \min_h \log \sigma^2 + \frac{1}{\sigma^2} \frac{1}{N} \sum_i (w^T x_i - y_i)^2,
\end{aligned}$$

where $x_i = (x_{i,1} \ x_{i,2} \ 1)^T$ and $w = (w_1 \ w_2 \ b)^T$. In the last expression, we observe that the summation term is independent from σ . Therefore,

$$w_{\text{MLE}} = \arg \min_w \sum_i (w^T x_i - y_i)^2,$$

which exactly corresponds to a *linear regression* problem. Then, we find w_{MLE} by canceling the gradient with respect to w , *i.e.*

$$\begin{aligned}
0 &= \nabla_w \sum_i (w^T x_i - y_i)^2 \\
&= \nabla_w \sum_i (w^T x_i - y_i)(w^T x_i - y_i) \\
&= \nabla_w \sum_i \left((w^T x_i)^2 + y_i^2 - 2w^T x_i y_i \right) \\
&= \nabla_w \left(w^T X^T X w + Y^T Y - 2w^T X^T Y \right) \\
&= 2X^T X w - 2X^T Y
\end{aligned}$$

where $X = (x_i^T) \in \mathbb{R}^{N \times 3}$ and $Y = (y_i) \in \mathbb{R}^N$. Finally, we have

$$\begin{aligned}
0 &= X^T X w_{\text{MLE}} - X^T Y \\
\Leftrightarrow w_{\text{MLE}} &= (X^T X)^{-1} X^T Y.
\end{aligned}$$

Afterwards, we find σ_{MLE} such that

$$\begin{aligned}
\sigma_{\text{MLE}} &= \arg \min_{\sigma} \log \sigma^2 + \frac{\text{MSE}}{\sigma^2} \\
&= \sqrt{\text{MSE}},
\end{aligned}$$

where MSE denotes the *mean squared error*

$$\frac{1}{N} \sum_i (w_{\text{MLE}}^T x_i - y_i)^2.$$

By (a), the fitted parameters estimate combinations of those of the network, $w_1 = a_3$, $w_2 = a_4 + a_5 a_2$, $b = a_5 \mu_3 + \mu_y$ and $\sigma^2 = a_5^2 \sigma_3^2 + \sigma_y^2$. Without observations of x_3 , the individual parameters $a_2, a_4, a_5, \mu_3, \mu_y, \sigma_3, \sigma_y$ cannot be recovered separately.

Exercise 9 Heteroscedastic linear regression

In linear regression with Gaussian noise of constant variance, $y_i \sim \mathcal{N}(w^T x_i, \sigma^2)$, the maximum likelihood estimate of the weights is $w_{\text{MLE}} = (X^T X)^{-1} X^T Y$, where the rows of X are the inputs x_i^T and $Y = (y_i)$. What does this estimate become if the noise is different for each sample, *i.e.* $y_i \sim \mathcal{N}(w^T x_i, \sigma_i^2)$ with known variances σ_i^2 ?

Solution. The log-likelihood of the weights is

$$L(w) = \sum_i \log \mathcal{N}(y_i | w^T x_i, \sigma_i^2) = - \sum_i \frac{(w^T x_i - y_i)^2}{2\sigma_i^2} + \text{const},$$

so that maximizing it minimizes a weighted sum of squares, each sample weighted by its precision $1/\sigma_i^2$. With $W = \text{diag}(1/\sigma_1^2, \dots, 1/\sigma_N^2)$,

$$\nabla_w L(w) = -X^T W(Xw - Y) = 0 \quad \Leftrightarrow \quad w_{\text{MLE}} = (X^T W X)^{-1} X^T W Y.$$

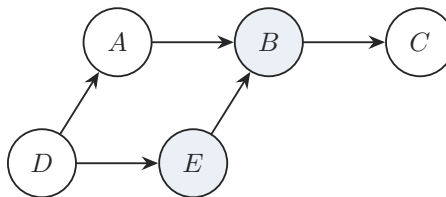
Noisy samples count less. With equal variances, W is proportional to the identity and the estimate of the homoscedastic case is recovered.

QUIZ

1. The Normal distribution $\mathcal{N}(\mu, \sigma)$ is described by the density function

- ☐ $p(x) = \frac{1}{\sigma} \exp(-(z + \exp(-z)))$, with $z = \frac{x-\mu}{\sigma}$.
- ☐ $p(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x-\mu|}{\sigma}\right)$.
- ☒ $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- ☐ $p(x) = 1/\pi\sigma \left[1 + \left(\frac{x-\mu}{\sigma}\right)^2\right]$.

2. Consider the Bayesian network shown below. We want to infer $\mathbf{P}(A | b, e)$ where A is the query variable, B and E are evidence variables, and C and D are hidden variables. Which of the following statements is true?



- ☐ $\mathbf{P}(A | b, e) \propto \sum_c \sum_d P(c | b) \mathbf{P}(b | A) P(b | e) \mathbf{P}(A | d) P(e | d) P(d)$
- ☐ $\mathbf{P}(A | b, e) \propto \sum_c \sum_d P(c | b) \mathbf{P}(b | A, e) \mathbf{P}(A | d) P(e | d)$
- ☒ $\mathbf{P}(A | b, e) \propto \sum_c \sum_d P(c | b) \mathbf{P}(b | A, e) \mathbf{P}(A | d) P(e | d) P(d)$
- ☐ $\mathbf{P}(A | b, e) \propto \sum_b \sum_e P(c | b) \mathbf{P}(A | b, e) \mathbf{P}(A | d) P(e | d) P(d)$

3. In a Bayesian network

- ☒ independence between two variables is guaranteed if all paths connecting them are inactive.
- ☐ dependence between two variables can be guaranteed using d-separation.
- ☐ a path between two nodes is active if it contains at least one active triple.
- ☐ a cascade triple is active if the center node is observed.

4. Consider a Bayesian network with a single binary variable, following a Bernoulli distribution. If we observe T positive realizations and F negative ones,
- ☐ the maximum likelihood estimate of the positive probability is $\frac{T}{F}$.
 - ☐ the maximum likelihood estimate of the positive probability is $\frac{T+F}{T}$.
 - ☒ the maximum likelihood estimate of the positive probability is $\frac{T}{T+F}$.
 - ☐ None of the above.